

Yale SCHOOL OF THE ENVIRONMENT

July 9, 2026

Dear Editor(s),

Please consider “Natural Capital and the Measurement of Productivity” for publication at AEJ: Macroeconomics. How best to account for the role of natural capital in the economy is a longstanding question. Recent (2025) changes to the System of National Accounts guidance along with “Beyond GDP” policy pushes globally have made understanding the role of natural capital in production especially salient. In the manuscript, we give a theoretical basis for how omitting natural capital biases estimates of total factor productivity (TFP) growth. Our estimates of the effects of additional natural capital measurement on the margin for a representative panel of 100 countries suggest that current practices that omit natural resources bias existing TFP estimates at scales that are important for budget forecasts and other long-run macroeconomic projections. The manuscript includes an appendix, intended for online-only publication. We recommend Vincenzo Quadrini from among the coeditors at AEJ: Macroeconomics.

The manuscript was previously submitted to AER:Insights as a short-form paper and rejected at the desk stage. We received feedback from Coeditor Prof. Gabriel Chodorow-Reich on how to sharpen the empirical section (Section 2) of the paper. Specific to coeditor Chodorow-Reich’s comments, Section 2 was lengthened substantially to more explicitly discuss the natural capital stocks we consider and outline more clearly how these stocks’ correlation with TFP growth evidences the reliance of production on nature we hypothesize. The paper has gone through several other substantial changes that were not motivated explicitly by the feedback from AER:I that we believe strengthen its contributions and make it suitable for submission as a full-length article at AEJ: Macroeconomics.

All data we use in this project are publicly available. All materials used as inputs to the manuscript, including raw data and computer code, will be made public at the time of publication and can be made available to either editors or referees if needed. The manuscript is not under review or consideration elsewhere. Please contact me with any further questions.

Sincerely,

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